Geometric Representation Theory

— *EYNTKA* Series —

Nathanael Chwojko-Srawley

February 17, 2026

Contents

1

Algebraic Groups

here

1.1 definitions

Recall that a *group scheme* is an S-scheme X for some base S along with morphisms $m : G \times_S G \to G$, $\iota : G \to G$, $e : \operatorname{Spec} S \to X$ that satisfy the group axioms; equivalently it is the group object in the category $\mathbf{Sch}_S$. If X is locally finite type over S, then X is called *locally algebraic*, and if it is furthermore quasicompact it is called *algebraic*. Combining these two definitions gets us what is common general definition of an algebraic group. In many cases, the definition is restricted to (abstract) varieties, which the following definition accommodates.

Definition 1.1.1: Algebraic Group

Let G be a variety or a group k-scheme. Then G is an *algebraic group* if it is finite type over k (which is automatic of G is a variety), and G has a group structure with multiplication and inversion:

$$\mu : G \times G \to G \qquad \text{and} \qquad \iota : G \to G$$

that are also morphisms of varieties (or k-schemes). The multiplication map μ is commonly written as $g \cdot h := \mu(g, h)$.

If we say G is an *algebraic group*, we shall assume it is an algebraic group over a *field*. If we want to be more general and not assume it is over a field, we shall say that G is an *algebraic group scheme*.

We shall mostly be working with groups over a perfect field or groups over an algebraically closed field of characteristic 0. By the Lefschetz principle (see [3, appendix]), if k is algebraically closed of characteristic 0, there is no theoretical issue assuming $k = \mathbb{C}$. We dropped the identity map as we can deduce it using the other two maps. To see why, consider

$$G \to G \qquad g \mapsto m(g, \iota(g)) = e$$

which is a (continuous) constant map. Restricting to the open affine $\operatorname{Spec} A \subseteq G$ containing e, we get a constant map $\varphi : \operatorname{Spec} A \to \operatorname{Spec} A$, $\varphi(x) = e$. The corresponding map $\varphi^{\#} : A \to A$ map every $\alpha \in A$ to the value of α at e, essentially $\alpha(e)$. These are all constants, hence $\operatorname{im}(\varphi^{\#}) \subseteq k$. We thus have a map $\varphi^{\#} : A \to k$. But then this means we have a map $\varphi : \operatorname{Spec} k \to \operatorname{Spec} A \hookrightarrow G$. Note how since k is assumed to be algebraically closed, by the correspondence of k-valued points and maximal points of a scheme, e must be closed. It is important that there *do* exist group-schemes where the constant is nonzero (see this Stacks Project example), however any group-scheme over a field necessarily has a closed identity.

Note that G is *not* a topological group (except in dimension 0) as $G \times G$ has the *Zariski topology* and not the product topology (show a topological group that is T_1 must be T_2, but G is not T_2). The translation map $x \mapsto xy$ is always an isomorphism, which also means that all geometric properties points are identical. As there are always smooth points, all points of G must be smooth.

Proposition 1.1.2: Properties of Algebraic Groups

Let G be an algebraic group over a field an algebraically closed field $k = \overline{k}$. Then:

1. G is separated.
2. (Cartier's Theorem) G is smooth.
3. If G is furthermore connected, then it is geometrically integral over k.
4. Only one irreducible component contains the identity.
5. G is equidimensional so that $\dim(G) = \dim_g(G)$ for all $g \in G$ (recall $\dim_g(G)$ is the dimension of g at G)
6. Closed subgroup of an algebraic group is an algebraic group
7. the direct product (really, fibered product) of algebraic groups is an algebraic group.

***Proof*:**

1. If $\Phi : G \times G \to G$ given by $(g, h) \to g \cdot h$, as e is closed, and $\Delta = \Phi^{-1}(e) = \{(g, g) \mid g \in G\}$, we have that the diagonal is closed and hence G is separated.
2. This is simply combining facts from [1], namely the module of differentials of G over k is free, G is smooth
3. (put the right reference)
4. Let's say $X = \bigcup_i^m X_i$ and $X_1, ..., X_n$ are (irreducible) components contain the identity. Then $X_1 X_2 \cdots X_n$ contains the identity. The product map $\prod_i^n X_i \to X$ maps irreducible

to irreducible (as is easy to check topologically) hence it must be that $X_1 X_2 \cdots X_n \subseteq X_i$ for some i. But certainly $X_j \subseteq X_1 X_2 \cdots X_n$ for each $1 \leq j \leq n$. Thus it must be that a single component contains the identity.

5. exercise.

6. exercise

7. exercise.

We give the component with the identity a name:

Definition 1.1.3: Identity Component

Let G be an algebraic group. Then the unique irreducible component containing the identity is called the *identity component*. It will be denoted G°.

Proposition 1.1.4: Identity Component and Connectedness

Let G be an algebraic group. Then:

1. $G^\circ \leqslant G$ is a normal subgroup of finite index in G whose cosets are connected and irreducible.

2. each closed subgroup of finite index in G contains G°.

***Proof*:**

1. For each $x \in G^\circ$, certainly $x^{-1}G^\circ$ is an irreducible component passing through e, hence $x^{-1}G^\circ = G^\circ$. The same logic shows $xG^\circ x^{-1} = G^\circ$. Therefore, $(G^\circ)^{-1} = G^\circ$ and $G^\circ G \circ = G^\circ$, showing G° is a normal subgroup. The cosets are all connected as they are all disjoint and translations of G° which is connected.

2. Let H be a closed subgroup of finite index. Then its compliment is also closed as its a finite union of closed sets, hence H is also open, hence connected. Thus the left cosets of H partition G°, and as G° is connected and meets H, $H \subseteq G^\circ$.

As the term irreducible in representation theory has a different meaning, we shall usually say that G is *connected* if $G = G^\circ$; the above makes it unambiguous as the irreducible components are cosets of G°, and if G is connected it can only have G° as it's single irreducible component.

The following will not be proven here, but it is important to know as it shows why the assumption that G is a variety is not much of a restriction

Theorem 1.1.5: Algebraic Group, then Quasi-Projective

Let G be an algebraic group over a field k. Then G is *quasi-projective* over k.

***Proof*:**
see https://stacks.math.columbia.edu/tag/0BF6.

It is important to take a moment to single out proper algebraic groups as they are *not* the focus of the book, and we must justify why:

Definition 1.1.6: Abelian Variety

Let k be a field. An *abelian variety* is a group scheme over k which is also a proper, geometrically integral variety over k.

There are many equivalent definitions of abelian variety (ex. proper for abelian variety implies projective, and geometrically integral shall imply smooth), but we leave this proof for a book focused on abelian varieties. Notice that they are not defined to be abelian, and yet their group structure shall *forced* to be abelian. Furthermore, as they are proper they have no non-constant global functions which we shall see implies the map $\rho : A \to \mathrm{GL}_n(k)$ must be *trivial*, namely their representation theory isn't of much intersect to this book[1].

The category of algebraic groups (or the group-objects in $\mathbf{Var}_k$ or $\mathbf{Sch}_k$) will have groups that are affine proper, or neither. However, due to a theorem by Chevalley, the study of algebraic group splits into "affine" and "abelian". Namely G has a normal subgroup H such that H is affine and G/H is projective (i.e. an "abelian variety").

Theorem 1.1.7: Chevalley Theorem

Let k be a perfect field and G an algebraic group over k. Then there exists a unique normal linear algebraic closed subgroup H in G for which G/H is an abelian variety. That is, there is a unique short exact sequence of algebraic groups

$$1 \to H \to G \to A \to 1$$

with H linear algebraic and A an abelian variety. The formation of H commutes with base change to an arbitrary perfect field extension over k.

***Proof*:**
see [4]

Thus, this book focuses on *affine algebraic groups.*

Lemma 1.1.8: Affine Algebraic Groups are Linear

Let G be an affine algebraic group over k. Then G is a closed subgroup of the affine algebraic group GL_n.

[1] This is "fixed" by studying the line bundles on A and their change under the Fourier transform (more specifically the Fourier-Mukai Transform); this recovers the notion of characters from the representation of finite abelian groups

***Proof*:**
Let $G = \text{Spec}\, A$ where A is a finitely generated k-algebra so that $k[x_1, ..., x_n]/(f_1, ..., f_m)$. Define a right-regular action of G on itself: $r_g(x) = xg$. This induces an action on $\Gamma(G)$, namely for any $f \in \Gamma(G)$:

$$g \cdot f(x) = f(xg)$$

giving a homomorphism $\rho_g : G \to \text{Aut}_k(A)$. If the codomain can be restricted to a finite-dimensional vector space, we have $\rho_g : G \to \text{GL}(V) \cong \text{GL}_n(k)$, giving us the desired (closed) embedding.

To that end, we first show that for any $f \in A$, it's orbit is a finite dimensional vector space. First, given $\mu : G \times_k G \to G$, by fineness we get the dual map $\mu^* : A \to A \otimes_k A$. For any $f \in A$,

$$\mu^*(f) = \sum_i^n f_i \otimes_k h_i$$

for some finite n and choices of f_i, h_i induced by μ. Then for any x, g:

$$(g \cdot f)(x) = f(xg) = \sum_{i=1}^m h_i(g) f_i(x)$$

And hence:

$$g \cdot f = \sum_{i=1}^m h_i(g) f_i$$

And so each $g \cdot f$ lies in the subspaces spanned by $\{f_1, ..., f_n\}$. If W_f is the space spanned by these, W_f is a finite dimensional k-vector space.

Now, as $\Gamma(G)$ is a finitely generated k-algebra, it is generated by $\alpha_1, ..., \alpha_m$. Each of these give a finite dimensional vector space W_{α_j}. Let:

$$V = \sum_i^m W_{\alpha_j}$$

Thus, the action of G restricts in the codomain to $\text{GL}(V)$; if $n = \dim(V)$ then we have:

$$\rho : G \to \text{GL}_n(k)$$

Now, it is easy to see that $\text{GL}_n(k)$ is an algebraic group as:

$$\text{GL}_n(k) \cong k[X_{11}, ..., X_{nn}, \overset{-1}{\det}]$$

Next, ρ is a k-morphism as the map $\rho^\# : \Gamma(\text{GL}_n(k)) \to \Gamma(G)$ is a k-algebra. To see this, let $v_1, ..., v_n$ be a basis for V. Then:

$$\rho(g) \cdot v_j = \sum_{i=1}^n M_{ij}(g) \cdot v_i$$

where $M_{ij} : G \to k$ are the appropriate functions that represent the matrix entries for the map $\rho(g)$. Then $\rho^\#(X_{ij}) = M_{ij}$, so certainly the map is k-regular. Furthermore, this shows that ρ is *surjective*, and hence $G \to \text{GL}_n(k)$ is a *closed immersion*, completing the proof.

Hence, we have reduced to showing that we are interested in groups G defined by a finite number of polynomial equations!

1.1.1 Examples of Algebraic Groups

Starting with the 0 dimensional example. All finite groups are affine algebraic groups. Indeed, every finite subgroup G is a subgroup of S_n, and S_n is a subgroup of $\mathrm{GL}_n(k)$ by associating a permutation $\sigma \in S_n$ to the matrix $I_\sigma \in \mathrm{GL}_n(k)$ where σ permutes the rows of the identity matrix $I \in \mathrm{GL}_n(k)$.

This also means that all finite groups can be interpreted as schemes. Given a finite group G interpreted as a closed algebraic subgroup of $\mathrm{GL}_n(k)$, then this scheme is isomorphic to $\operatorname{Spec}\left(\prod_{i=1}^{|G|} k\right)$. This is the *constant finite algebraic groups.*

Interestingly, it is *not* he case that finite group schemes are bijective correspondence with abstract groups; there are *more* group schemes. To see this, consider the group $G = \mu_3$ which are the 3rd roots of unity. Purely on the level of group theory, $\mu_3 \cong C_3$ (the cyclic group of order 3). However, μ_3 has further structure. Take the $\mathbb{Q}$-scheme $G_{\mathbb{Q}} = \operatorname{Spec}\mathbb{Q}[x]/(x^3-1)$. Then $G_{\mathbb{Q}}$ has 2 points as $\mathbb{Q}[x]/(x^3-1) \cong \mathbb{Q}\times\mathbb{Q}(\omega)$ given a 3rd root of unity ω. In this group, we "fused" two points (namely ω, ω^2). To recover G, we must look at the *geometric points* of $G_{\mathbb{Q}}$, namely the algebraic closure (i.e. base-changing to $\overline{\mathbb{Q}}$) shall separate these points so that $G_{\overline{\mathbb{Q}}}$ has three points. We once again have recovered C_3, but now there is a natural action of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ associated to C_3! We can interpret the $\mathbb{Q}$-valued points, $G_{\overline{\mathbb{Q}}}(\mathbb{Q})$, as the fibers of $G_{\overline{\mathbb{Q}}}$ which explains why they "split" when extending to $\overline{\mathbb{Q}}$.

More generally, if G is a finite étale algebraic group over k, then $\Gamma(G)$ is an étale algebra, that is:

$$\Gamma(G) \cong L_1 \times \cdots \times L_r$$

Any such scheme scheme corresponds to a finite group along with a (continuous) action of $\mathrm{Gal}(\overline{k}/k)$. These are called *twisted finite algebraic groups.* In the extreme case, we have some finite extension L/K of degree n so that $G_k = \operatorname{Spec}(L)$ has a single point, but n geometric points that are twisted via $\mathrm{Gal}(\overline{K}/K)$ where the action naturally restricts to $\mathrm{Gal}(L/K)$.

The term "twisted" comes directly from Galois cohomology and shares the same geometric intuition as twisted vector bundles. A finite group scheme G over k is called a *twisted form* of a constant group scheme G_{const} if they become isomorphic over the separable algebraic closure $\overline{k}$. That is, there exists an isomorphism of group schemes $\varphi : G_{\overline{k}} \xrightarrow{\sim} (G_{const})_{\overline{k}}$. While G and G_{const} look identical geometrically over $\overline{k}$, the isomorphism φ fails to commute with the natural Galois actions on both sides. This failure is measured by the map $a_\sigma = \varphi \circ \sigma \circ \varphi^{-1} \circ \sigma^{-1}$, which defines a 1-cocycle in $H^1(\mathrm{Gal}(\overline{k}/k), \mathrm{Aut}(G_{const}))$. This classification of twisted forms by H^1 is identical to the classification of varieties described in [2]

We can interpret this physically as a conflict between two Galois actions. The constant group G_{const} possesses a "standard" action σ_{std} that acts trivially on the group elements (simply permuting the copies of k). The twisted group G possesses its own intrinsic Galois action derived from its definition over k. The cocycle a_σ acts as a set of instructions for "gluing" the standard object to itself with a twist: to reconstruct the group G from G_{const}, we define a *new* action $\sigma_{new}(x) := a_\sigma(\sigma_{std}(x))$ on the geometric points. The k-rational points of the twisted group G are exactly the fixed points of this new, twisted action.

Returning to our example of $G = \mu_3$ and $G_{const} = C_3$, over $\overline{\mathbb{Q}}$ we have an isomorphism mapping the roots $\{1, \omega, \omega^2\}$ to the abstract elements $\{e_1, e_\sigma, e_{\sigma^2}\}$. The standard action on C_3 fixes every element. However, the Galois action on μ_3 (complex conjugation τ) swaps ω and ω^2. This mismatch means our cocycle $a_\tau \in \mathrm{Aut}(C_3)$ must be the automorphism that swaps e_σ and e_{σ^2}. Thus, while the constant group scheme corresponds to the trivial class in H^1, the roots of unity μ_3 correspond to a non-trivial class in $H^1(\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}), \mathrm{Aut}(C_3))$, representing a specific twist of the constant structure.

For those familiar with group schemes as group functors we can generalize this idea using functor of points. If $G = \mathrm{Spec}\,(A)$ is an affine algebraic group take the representable functor:

$$R \mapsto G(R) = \mathrm{Hom}_{k\text{-}\mathbf{Alg}}(A, R)$$

For constant group schemes we have $G(k) = \mathrm{Hom}(k^n, k)$. Since maps to k must pick out one component, there are exactly n such maps. For $G = \mathrm{Spec}\,(L)$, then $G(k) = \mathrm{Hom}(L, k)$ which has only the identity as L only has the trivial map into k (i.e. $G(k)$ only see's the identity, or everything is glued to the identity). If instead we take L-points: $G(L) = \mathrm{Hom}(L, L) = \mathrm{Gal}(L/k)$, recovering the twisting information.

Next, consider the following 1-dimensional examples:

1. $G = \mathbb{G}_a = \mathbb{A}^1$ so that the map is given by $\mu(x, y) = x + y$, $\iota(x) = -x$ and $e = 0$.
2. $G = \mathbb{G}_m = (\mathbb{A}^1)^*$ is given by $\mu(x, y) = xy$, $\iota(x) = x^{-1}$ and $e = 1$.

Both of these groups are irreducible and 1-dimensional. We shall see in ref:HERE that (up to isomorphism) these are the *only* irreducible 1-dimensional groups.

Dimension 2 and higher start having non-trivial examples, and so they are left to be treated by the book.

A natural generalization of $\mathbb{G}_a$ is $\mathbb{A}^n$ with its natural point-wise additive structure. We can denote this $\mathbb{G}_a(\mathbb{A}^n)$. We cannot do the same for $(\mathbb{A}^n)^*$ as there is no inverse (ex. $(1, 0, 1)$ has no multiplicative inverse). Instead, we define the *maximal torus* $\mathbb{G}_m(\mathbb{A}^n) = (\mathbb{A} \setminus \{0\})^n$.

As all algebraic groups are linear, it is important look at importnat subgroup of the general linear group. Let K be a field. Then $\mathrm{GL}_n(K)$ is an open subset of $M_n(K)$ given by the non-vanishing set of $\det(T_{ij})$ for appropriate functions T_{ij}, and hence is an affine algebraic group. Note that the orthogonal group O_n is a natural non-connected algebraic group.

The following algebraic groups will repeatedly be brought up:

1. The special linear group $\mathrm{SL}_n(K)$, given by $\det(T_{ij}) - 1$
2. The symplectic group $\mathrm{Sp}_{2n}(K)$ consisting of all $x \in \mathrm{GL}_{2n}(K)$ where
$$ {}^t x \begin{pmatrix} 0 & J \\ -J & 0 \end{pmatrix} x = \begin{pmatrix} 0 & J \\ -J & 0 \end{pmatrix}$$
where ${}^t x$ is the transpose.
3. The special orthogonal group $\mathrm{SO}_{2n+1}(K)$ (where $\mathrm{char} K \neq 2$) given by all $x \in \mathrm{SL}_{2n+1}(K)$ where:
$$ {}^t x s x = s$$
where
$$s = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & J \\ 0 & J & 0 \end{pmatrix}$$

We shall sometimes specify the special orthogonal group $\mathrm{SO}_n(K)$ where $s = \begin{pmatrix} 0 & J \\ J & 0 \end{pmatrix}$

Next, let us introduce two more subgroups groups of GL_n that shall be important for their role in forming a composition series that any algebraic group G decomposes into. Namely, we have a natural maximal subnormal series:

$$e \trianglelefteq U \trianglelefteq N \trianglelefteq A \trianglelefteq G^\circ \trianglelefteq G$$

where each G_i/G_{i+1} is simple and:

1. G contains a unique connected normal subgroup G° such that G/G° is finite (see proposition 1.1.4)

2. G° contains a largest (hence unique) connected afine normal subgroup A such that G°/A is an abelian variety (see theorem 1.1.7)

3. A contains a largest (hence unique) connected affine solvable normal subgroup N such that A/N is semisimple (see ref:HERE)

4. N contains a largest (hence unique) connected unipotent normal subgroup U such that N/U is a torus (see ref:HERE)

Thus, the *torus* and *unitpotent* groups are singled out for their general immediate usefulness for upcoming examples:

Definition 1.1.9: [Algebraic] Torus

Let $\mathrm{GL}(V)_k$ be the affine general linear group over a finite dimensional k-vector space V and $T \subseteq \mathrm{GL}(V)$ an affine subgroup. Let $Z = Z(\mathrm{GL}(V))$ be the center of $\mathrm{GL}(V)$ (i.e. all diagonal matrices). Then T is said to be of *multiplicative type* if when extending to $\overline{k}$ $T_{\overline{k}} \subseteq Z_{\overline{k}}$. In other words, there exists a basis for V such that $T_{\overline{k}}$ is represented by diagonal matries, i.e. the elements of an algebraic torus are semisimple endomorphisms of V.

If T is furthemore connected, it is called a *torus.*

The name comes from the fact that over the complex numbers $\mathbb{C}$, the group of points of an algebraic torus is isomorphic to a subgroup of $(\mathbb{C}^\times)^n$, which contains the standard topological torus $(S^1)^n$ as its maximal compact subgroup.

Definition 1.1.10: Unipotent Group

Let $\mathrm{GL}(V)_k$ be the affine general linear group over a finite dimensional k-vector space V and $G \subseteq \mathrm{GL}(V)$ an affine subgroup. Then G is called *unipotent* if there exists a basis of V relative to which G is contained in the group U_n of all $n \times n$ matrices of the form

$$\begin{pmatrix} 1 & * & \cdots & * & * \\ 0 & 1 & \cdots & * & * \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & * \\ 0 & 0 & \cdots & 0 & 1 \end{pmatrix}$$

In particular, the elements of a unipotent group are unipotent endomorphisms of V.

I think I should metnion solvable, reducitve, simple, quasi-simple, semi-simple, later

1.1.2 Hopf Algebra

We'll focus on affine algebraic groups so $G = \operatorname{Spec} A$. So that $G \times G \to G$ corresponds to $\Delta : A \to A \otimes A$ that makes A into a *Hopf algebra*.

1. Δ is co-associative
2. there exists a co-unit $\epsilon : A \to k$ satisfying the usual axiom of a ring, and
3. there exists an inversion $\iota : A \to A$.

1.2 *Algebraic Groups as Functors

When picking $\mathrm{GL}_n(R)$, there is a choice R. We can eliminate this choice by capturing the required properties functorially. Let $S \subseteq k[x_1, ..., x_n]$ be a set of polynomials and R a k-algebra. Then:

$$S(R) = \{(a_1, ..., a_n) = a \in R^n \mid f(a) = 0 \forall f \in S\}$$

given a k-homomorphism $R \to R'$, we get a map $S \to S'$, and hence a functor:

$$\mathbf{Alg}_k \to \mathbf{Set}$$

We can thus think of a choice of such a functor as being an algebraic set. Naturally, if we instead take functors:

$$\mathbf{Alg}_k \to \mathbf{Grp}$$

we call such a functor an algebraic group. For example, we can define the functor $\mathrm{SL}_n : \mathbf{Alg}_k \to \mathbf{Grp}$ where any $R \in \mathbf{Alg}_k$ maps to $\mathrm{SL}_n(R)$ that are all elements that satisfy the equations $\det(x_{ij}) - 1$ with the determinant defined as:

$$\det(x_{ij}) = \sum_{\sigma \in S_n} |\sigma| X_{1\sigma(1)} \cdots X_{n\sigma(n)} \in k[x_{11}, ..., x_{nn}]$$

As the functor needs to be defined by polynomials, this excludes some behaviours, for example you can't have "indicator" functions such as:

$$F(R) = \begin{cases} (k,+) & R = k \\ e & \text{otherwise} \end{cases}$$

Let us now look at affine algebraic group.s Recall that if k is Noetherian and $k[x_1, ..., x_n]$ is a k-algebra, then every ideal is finitely generated (Hilbert Basis Theorem, see [3]). Thus an affine algebraic group (seen as a functor) is isomorphic to a functor defined by a finite set of polynomials. Next, let $A = k[x_1, ..., x_n]/I$ be a k-algebra and recall that the map $A \to R$ can be thought of as evaluating the polynomials in A at R-points. Therefore the functor $R \mapsto I(R)$ sending a k-algebra R to the zero-set of a in R^n is canonically isomorphic to the functor:

$$R \mapsto \mathrm{Hom}_{\mathbf{Alg}_k}(A, R)$$

Thus, algebraic group are representable functors. Now consider a k-algebra A together with a k-algebra homomorphism $\Delta : A \to A \otimes A$. For any k-algebra R, the map

$$f_1, f_2 \mapsto f_1 \cdot f_2 \stackrel{\text{def}}{=} (f_1, f_2) \circ \Delta : h^A(R) \times h^A(R) \to h^A(R),$$

is a binary operation on $h^A(R)$, which is natural in R. Therefore, we the diagonal map captures the multiplication structure and we can write an algebraic group as a pair (G, Δ) .

(Milne has a quick word here about augmenting a algebraic group with a $\mathbb{Z}/2\mathbb{Z}$-grading to get an affine supergroup which are important in supersymmetry).

Index

Bibliography

[1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Geometry*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algGeo.pdf.

[2] Nathanael Chwojko-Srawley. *Everything You Need To Know About Galois Cohomology*. EYNTKA. 2025. URL: https://nathanaelsrawley.com/assets/pdfs/books/EYNTKA_galois_cohomology.pdf.

[3] Nathanael Chwojko-Srawley. *Everything You Need To Know About Undergraduate Algebra*. 1st ed. 2026. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebra.pdf.

[4] Brian Conrad. “A modern proof of Chevalley’s theorem on algebraic groups”. In: 17 (2002).